WORKED SOLUTION

STEP III 2013, Question 2

Pure • Maclaurin Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q2

The key idea

Clearing the square root first, by writing $y\sqrt{1-x^2} = \arcsin x$, makes the first derivative painless. The higher-order relation then comes from differentiating that first-order equation repeatedly with Leibniz's rule. At $x = 0$ most of the terms vanish and we are left with a two-step recurrence for the derivatives, which builds the Maclaurin coefficients one at a time. The final sum looks like the series evaluated at a well-chosen value of x.

Solution

The first-order equation

Multiply up to get $y\sqrt{1-x^2} = \arcsin x$ and differentiate both sides with respect to x, using the product rule on the left:

$$\sqrt{1-x^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{xy}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}}.$$

Multiplying through by $\sqrt{1-x^2}$ (non-zero for $|x| < 1$) gives

$$(1-x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - xy - 1 = 0,$$

as required.

The higher-order relation

Write $y^{(k)}$ for $\dfrac{\mathrm{d}^k y}{\mathrm{d}x^k}$. Differentiate the first-order equation $n+1$ times. The constant 1 disappears, and Leibniz's rule handles the two products. Since $1-x^2$ has derivatives

$-2x$, -2 and then 0, and x has derivatives 1 and then 0,

$$\frac{\mathrm{d}^{n+1}}{\mathrm{d}x^{n+1}}\Big[(1-x^2)y'\Big] = (1-x^2)y^{(n+2)} + \binom{n+1}{1}(-2x)\,y^{(n+1)} + \binom{n+1}{2}(-2)\,y^{(n)}$$
$$= (1-x^2)y^{(n+2)} - 2(n+1)x\,y^{(n+1)} - n(n+1)\,y^{(n)},$$

$$\frac{\mathrm{d}^{n+1}}{\mathrm{d}x^{n+1}}\Big[xy\Big] = x\,y^{(n+1)} + (n+1)\,y^{(n)}.$$

Subtracting the second from the first and collecting terms,

$$(1-x^2)y^{(n+2)} - (2(n+1)+1)x\,y^{(n+1)} - (n(n+1)+(n+1))y^{(n)} = 0,$$

that is,

$$(1-x^2)y^{(n+2)} - (2n+3)x\,y^{(n+1)} - (n+1)^2 y^{(n)} = 0.$$

Nothing here needed $n \geq 1$: differentiating once ($n = 0$) gives the same pattern, $(1-x^2)y'' - 3xy' - y = 0$, and we will use that case too.

The Maclaurin series

Put $x = 0$. The middle term carries a factor x, so it drops out and

$$y^{(n+2)}(0) = (n+1)^2\,y^{(n)}(0) \qquad (n \geq 0).$$

The recurrence moves in steps of two, so we need two starting values. From the definition, $y(0) = \arcsin 0 = 0$, and the first-order equation at $x = 0$ gives $y'(0) - 0 - 1 = 0$, so $y'(0) = 1$.

? **Ask yourself** How do we get every derivative from just these two values?

The recurrence links $y^{(n)}(0)$ only to $y^{(n+2)}(0)$, so the even derivatives form one chain started by $y(0)$ and the odd derivatives form another started by $y'(0)$.

Even powers. Since $y(0) = 0$, the recurrence gives $y''(0) = 1^2 \cdot 0 = 0$, then $y^{(4)}(0) = 0$, and so on: $y^{(2k)}(0) = 0$ for every k. So the coefficient of every even power of x is zero. (This fits with y being an odd function, as a quotient of the odd function $\arcsin x$ by the even function $\sqrt{1-x^2}$.)

Odd powers. Using the recurrence with $n = 1, 3, \ldots, 2k-1$,

$$y^{(2k+1)}(0) = (2k)^2(2k-2)^2 \cdots 4^2 \cdot 2^2 \cdot y'(0) = (2 \cdot 4 \cdots 2k)^2 = (2^k\,k!)^2,$$

which also holds for $k = 0$. Dividing by $(2k+1)!$ gives the Maclaurin coefficients:

$$\frac{\arcsin x}{\sqrt{1-x^2}} = \sum_{k=0}^{\infty} \frac{2^{2k}(k!)^2}{(2k+1)!}\,x^{2k+1} = x + \frac{2^2}{3!}x^3 + \frac{2^2 \cdot 4^2}{5!}x^5 + \cdots,$$

so the general odd term is $\dfrac{2^{2k}(k!)^2}{(2k+1)!}x^{2k+1}$, and the general even term is $0 \cdot x^{2k}$.

The infinite sum

The nth term of the sum is $\dfrac{(n!)^2}{(2n+1)!}$, with $n = 0$ giving the leading 1. Comparing with the series, we want to remove the factor 2^{2n}, so choose x with $2^{2n}x^{2n+1}$ a constant multiple of 1: $x = \frac{1}{2}$ gives $2^{2n}(\frac{1}{2})^{2n+1} = \frac{1}{2}$. Hence

$$\frac{\arcsin\frac{1}{2}}{\sqrt{1-\frac{1}{4}}} = \frac{1}{2}\sum_{n=0}^{\infty}\frac{(n!)^2}{(2n+1)!}.$$

The left-hand side is $\dfrac{\pi/6}{\sqrt{3}/2} = \dfrac{\pi}{3\sqrt{3}}$, so

$$1 + \frac{1}{3!} + \frac{2^2}{5!} + \frac{2^2\times 3^2}{7!} + \cdots = \frac{2\pi}{3\sqrt{3}} = \frac{2\sqrt{3}\,\pi}{9}.$$

Checks

- Multiplying the known series $\arcsin x = x + \frac{1}{6}x^3 + \cdots$ by $(1-x^2)^{-1/2} = 1 + \frac{1}{2}x^2 + \cdots$ gives $x + (\frac{1}{6}+\frac{1}{2})x^3 + \cdots = x + \frac{2}{3}x^3 + \cdots$, and indeed $\frac{2^2}{3!} = \frac{2}{3}$.
- The first five terms of the sum are $1 + 0.16667 + 0.03333 + 0.00714 + 0.00159 \approx 1.2087$, already close to $\frac{2\pi}{3\sqrt{3}} \approx 1.20920$; the terms shrink by a factor of about 4 each time, so the remainder is small.

Common slips

- Forgetting the binomial coefficient $\binom{n+1}{2}$ on the $y^{(n)}$ term in Leibniz's rule, which spoils the $(n+1)^2$.
- Using the recurrence only for $n \geq 1$ and so never establishing $y''(0) = 0$; the case $n = 0$ (one differentiation) is needed for the even chain.
- Substituting $x = 1$ or $x = \frac{1}{4}$ in the final part: the factor 2^{2n} has to be cancelled by x^{2n}, which forces $x = \frac{1}{2}$.

Beyond STEP

Here we "ignored questions of convergence", but a first analysis course makes this precise: the series has radius of convergence 1 by the ratio test, and Taylor's theorem with remainder shows it really does equal the function inside that interval. The coefficients $\frac{2^{2k}(k!)^2}{(2k+1)!}$ are closely related to central binomial coefficients, and the same function generates the series for $(\arcsin x)^2$ after one integration. For the analysis behind power series, see Abbott, *Understanding Analysis*.

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